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Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

EE7204 / EC7205

Computer Vision–Based Elder Monitoring System

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Contents

Contents	1
1 Problem Statement	2
2 Objectives	3
3 Literature Review	4
3.1 Vision-based Human Fall Detection Systems using Deep Learning (2022)	4
3.2 Development of Artificial Intelligence Edge Computing-Based Wearable Device for Detection and Prevention of Fall of Elderly (2024)	4
3.3 ElderFallGuard: Real-Time IoT and Computer Vision-Based Fall Detection System for Elderly Safety (2025)	4
3.4 Gaps in Literature	5
4 Methodology	6
4.1 System Architecture Overview	6
4.2 Hardware Components	6
4.3 Software Stack	7
4.4 Dataset Preprocessing	7
4.5 Model Selection and Architecture	7
4.6 Model Training and Fine-tuning	8
4.7 Model Quantization	8
4.8 Pose Estimation Integration	8
4.9 Classification and Decision System	9
4.10 Evaluation Methodology	9
5 Dataset	10
6 Expected Outcomes	11
7 Percentage of Image Processing & Computer Vision Involvement	12

1. Problem Statement

Elder people's falls and unsafe walking accidents are a serious public health concern. As a result of that, they face injuries, Long-term disability, and higher healthcare costs. Wearable sensors and manual supervision are examples of traditional fall monitoring solutions that have drawbacks like user non-compliance, discomfort, battery dependency, and slow reaction times. Many current methods depend on cloud-based processing or high-computation hardware, vision-based monitoring systems have emerged as a potential invasive alternative that not appropriate for low-cost , real time home environment deployment. Furthermore, Limited attention has been given to optimizing deep learning models for resource-constrained edge devices to maintain detection accuracy and privacy.

This project creates a computer vision-based elderly monitoring system that uses pretrained deep learning models that have been optimized and quantized for effective Raspberry Pi execution to identify fall-related risks (such as falls on stairs or prolonged inactivity). The suggested method aims to provide real-time, privacy-preserving elder protection suitable in real-world home settings. [1,2].

2. Objectives

- developing a computer vision-based system that uses motion analysis and pose estimation to identify elderly falls and safety hazards in real time.
- optimize for edge deployment by fine-tuning pretrained deep learning models for reliable fall detection on small datasets.
- the models will enable effective operation on Raspberry Pi hardware by lowering computational requirements.
- improve elderly safety by incorporating guardian alerting systems when dangerous events are detected.
- evaluate the accuracy, speed, and power consumption of the system in home environment simulations.
- optimize for edge deployment by fine-tuning pretrained deep learning models for reliable fall detection on small datasets.
- the models will enable effective operation on Raspberry Pi hardware by lowering computational requirements.
- improve elderly safety by incorporating guardian alerting systems when dangerous events are detected.
- evaluate the accuracy, speed, and power consumption of the system in home environment simulations.

3. Literature Review

3.1. Vision-based Human Fall Detection Systems using Deep Learning (2022)

In vision-based Human Fall Detection systems using Deep Learning(2022),the authors had provided a comprehensive review on deep learning approach used in vision-based fall detection. This had categorized existing methods into 2D CNNs, 3D CNNs, and hybrid architectures. This review demonstrate that the widespread use of transfer learning, particularly with pre-trained models such as VGG-16 and ResNet. Those are effectively extracted spatiotemporal features from video sequences.Which reduced the need for larger labeled datasets. The results indicate that several models achieved high performance, with accuracies over 98% on benchmark datasets such as URFD.This was especially for under controlled experimental conditions.This review highlighted the key limitation when implementing real world deployment,including to lighting variations, occlusions and high computation cost required.The authors observe that a growing trend toward privacy-aware edge AI approaches. At the same time noted that lack of research on optimization techniques, such as model quantization for devices like Raspberry Pi Which is remained as underexplored. [3]

3.2. Development of Artificial Intelligence Edge Computing–Based Wearable Device for Detection and Prevention of Fall of Elderly (2024)

The study Development of Artificial Intelligence Edge Computing–Based Wearable Device for Detection and Prevention of Fall of Elderly (2024) presents an AI intergrated wearable system that applies to edge computing to fall detection and prevension. Although this approach is mainly sensor-based and integrates computer vision by processing on-device camara data with lightweight CNN models.Which trained on fall-related datasets.The system achieves around 95% detection accuracy with a latency below one second.That supported by INT8 quantization for better energy efficiency. The main strength of the system is its hybrid design which has made of combining camera and sensor inputs, that ensure reliable detection and real time alerts. However depending on wearable devices may reduces usability for some elderly individuals. That cause to comfort and compliance issues. In addition when deployment the system it causes some issues in complex environment such as stairs [4].

3.3. ElderFallGuard: Real-Time IoT and Computer Vision-Based Fall Detection System for Elderly Safety (2025)

ElderFallGuard: Real-Time IoT and Computer Vision-Based Fall Detection System for Elderly Safety (Riahi et al., 2025). This research introduces a "non-invasive IoT solution for efficient fall detection of elderly people in real-time using MediaPipe Pose estimation on regular RGB camera feed from a singular camera source. The technique involves calculating 33 body keypoints and using threshold-based rules to identify abnormalities. These

abnormalities include maintaining a prone state (horizontal orientation of body) longer than 3 seconds and low motions' variance longer than 2 seconds to finally determine a fall. No heavy deep learning classifier is used for the final decision; instead, rule-based logic processes keypoints, with fine-tuning applied to pose mapping on custom/simulated datasets including falls and activities of daily living (ADLs). The study indicates outstanding results on unseen datasets where accuracy, precision, recall, and f-score are 100% in a controlled environment. However, around 97% precision in a realistic environment. IoT helps in notifying care personnel in real-time via a telegram using timestamps and low-resolution camera snapshots. IoT connection allows for immediate notification of caregivers via Telegram, complete with timestamp and optional low-resolution images. The benefits of the proposed system include complete on-device processing for enhanced security, low CPU requirements since MediaPipe is CPU-friendly, and support for immediate response and smart home scenarios without the need for wearables. The system responds to non-compliance among the elderly by using wearables and immediately sends messages to minimize "time on ground" in case of a fall [5].

3.4. Gaps in Literature

By considering three studies highlights that complementary yet distinct limitations in vision based and edge enabled elderly fall detection. The 2022 review provides a broad overview of deep learning methods (2D/3D CNNs, hybrids with transfer learning from VGG-16/ResNet), achieving high accuracies but had identified the main barriers. Those are sensitivity to real world factors like lighting variations and occlusions, high computational costs and notably underexplored optimization for low power devices like raspberry pi. The 2024 study by Paramsivam advance edge efficiency with light-weight CNN hybrids and INT8 quantization, plus hybrid sensor vision fusion for reliable alerts but its wearable focus reduces usability due to compliance issues among elders. It struggles in complex environment like stairs where sensor visual integration falters. ElderFallGuard (Riahi et al., 2025) shifts to a fully non-wearable, privacy-preserving vision-only approach using MediaPipe Pose with simple threshold rules. It is delivering near-perfect results (100% controlled, 97% realistic) and IoT alerts (Telegram), yet it depends on static thresholds vulnerable to false positives in cluttered scenes and lacks dynamic motion tracking or quantization for ultra-low-power hardware. Collectively, these works show progress in accuracy, privacy, and edge processing but lack integration of robust motion analysis (e.g., optical flow), full quantization for Raspberry Pi-like devices, and overhead-view adaptations in a non-wearable, pure-vision system gaps your "Elder Watch" project addresses through YOLO-MediaPipe hybrid detection, optical flow enhancement, INT8 optimization, and ceiling-mounted setup for real-world robustness and affordability [3] [4] [5].

4. Methodology

4.1. System Architecture Overview

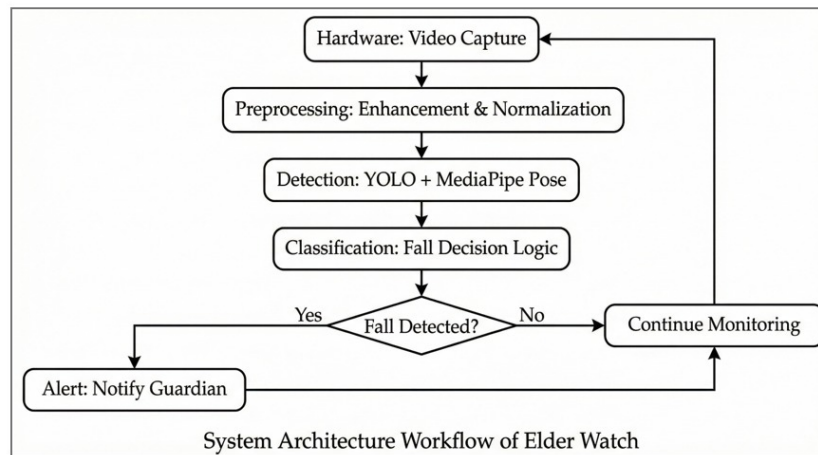


Figure 1: System Architecture Diagram4

Elder Watch's architecture follows a layered structure for effective and non-invasive monitoring. The hardware layer involves collecting video data with the help of a Raspberry Pi and camera module. A preprocessing stage refines and normalizes the data for effective analysis. The detection stage utilizes YOLO for locating people and MediaPipe for understanding the posture and movements of people. The classification/decision stage involves rule-based decision-making with the help of these parameters. The alert system then sends notifications to the guardians. The architecture ensures effective and resourceful monitoring, with a focus on effective and robust testing. Below diagram shows Full Architectural workflow.

4.2. Hardware Components

The hardware components are designed with the Raspberry Pi 5 [6] as the basis. The Raspberry Pi 5 [6] has a quad-core Arm Cortex A76 CPU running at a 2.4 GHz frequency and can support a maximum of 4 GB RAM. These components are adequate and provide the computational power required by the system to perform the inference in a timely manner. At the same time, the components are cost-effective, with the Raspberry Pi 5. The Raspberry Pi camera module 3 has a wide-angle view of about 120 degrees, which is adequate to cover the area above the staircase at a height of 2-3 meters. These components are connected with a 64 GB microSD storage device and a 5V and 5A power supply. There is an option to use a Google Coral USB accelerator, which can aid the system in performing the operations. However, the components are adequate and can support the operations. The components are designed to support the feasibility of the system and can support the deployment of the system in a home environment. The components are adequate and can support the deployment of the system in a home environment in a developing country like Sri Lanka.

4.3. Software Stack

The software stack is set up in such a manner as to provide the best edge efficiency. Raspberry Pi OS (64-bit Bookworm) is the base OS used. This OS provides support for the ARM architecture. TensorFlow Lite is being used to deploy the models in the quantized form. This allows for the deployment of lightweight models. OpenCV is being used for video processing and the calculation of the optical flow. Real-time pose estimation is being performed with the aid of the MediaPipe Pose library. To train the models, the models are being fine-tuned on the desktop GPU environment with the aid of the PyTorch and TensorFlow libraries. Python is being used as the primary programming environment. NumPy and PIL are being used in conjunction with Python for the numerical and image processing tasks. Twilio is another library being used in the stack for the provision of the alert functionality.

4.4. Dataset Preprocessing

The preprocessing starts with the extraction of the frames at 30 FPS with the help of the OpenCV library. This is followed by resizing the images into 640×640 or 416×416 dimensions, as appropriate, with the pixel values being normalized in the range 0,1. There are more data augmentations like rotate in clockwise and anticlockwise directions. The data is divided into train, validation, and test sets according to 70 percent, 15 percent and 15 percent. This process ensures the quality of the data while being able to withstand changes in illumination conditions or the orientation of the images. By following the existing benchmarks, the effectiveness of the proposed model can be validated.

4.5. Model Selection and Architecture

YOLOv8 Nano was chosen for the detection task because of its small model size, measured at 3.2 million parameters, which enables real-time inference and is efficient for edge devices. Moreover, it achieves a mean average precision rate greater than 90% for fall detection tasks. The architecture is based on a CSP Darknet backbone and C2F modules for efficient feature extraction, a PANet neck for multi-scale feature aggregation, and a decoupled head for bounding box regression and classification. The architecture is trained on $640 \times 640 \times 3$ RGB images, and the output is a detection classifying between people and falls, along with a corresponding confidence measure. As an alternative to YOLOv8 Nano, YOLOv11n will also be considered for benchmarking, which is scheduled for a 2024 release and is also a lightweight architecture with efficiency improvements through new optimizations. MediaPipe Pose is complementary to YOLO, as it is used for pose estimation, allowing for the detection of 33 body landmarks on the CPU in real-time. It uses BlazePose topology to accurately determine the keypoints, which is beneficial for viewing the postures of the subjects from the top. The process of integration follows a sequence where YOLO is used to detect and crop "person" instances where confidence levels are above 0.5, and then MediaPipe is utilized to detect keypoints to identify fall

indicators such as body orientation. The combination of these two models has proven to give more accurate results than using a single model in a cluttered environment. The models have been chosen based on their open-source support to ensure ease of tuning to suit the elderly environment.

4.6. Model Training and Fine-tuning

Training is carried out in two distinct phases. In Phase 1, a desktop setup is utilized with an NVIDIA GPU and the PyTorch library's Ultralytics library. The YOLOv8 [2] is deployed on the system. The weights of the model are initialized with pre-trained weights from the COCO dataset. The model is then fine-tuned on the KFall, CAUCAFall, and FDD datasets. The label is set to "person" and "fall." Phase 2 involves optimization via exporting the model to ONNX, then transforming the model into TensorFlow, followed by model optimization via the process of quantization. On the other hand, the evaluation metrics are mean average precision at 0.5 IoU precision, recall, F1 score, inference time, and model size. Overall, the approach involves the application of the technique known as transfer learning, which enables the application of general object detection models that are adapted to the specific needs related to falling, hence minimizing the need for data while maintaining high precision. MediaPipe is used as is, with some modifications on the back end.

4.7. Model Quantization

The model size is reduced using the quantization approach, which helps in reducing the model size for deployment on the Raspberry Pi using TensorFlow Lite. The full workflow go through below process. Post Training Quantization (PTQ) is the preferred method, with weight only quantization converting FP32 to INT8 and delivering a reduction in model size of up to four times with negligible impact to model accuracy. Integer quantization involves quantizing both activations and weights and requires a calibration set consisting of 500 to 1000 samples (various falls and Activities of Daily Living from the training set). If feasible, Quantization-Aware Training (QAT) can be used to simulate the effects of quantization during the fine-tuning stage, which is known to be more accurate but longer in duration. The expected benefits would be the reduction in size from the initial 25 MB FP32 model to a size of 6-7 MB in INT8, a 2 or 3X improvement in inference acceleration (e.g., from 50ms to 20ms per frame), along with a drop in accuracy of less than 2% based on validation datasets. Optimizes the code for running in real-time (≈ 30 ms/frame) on the Raspberry Pi 5's ARM CPU, an important consideration when monitoring the elderly since response times may be impacted [7].

4.8. Pose Estimation Integration

The process of integration begins with YOLO detecting human bounding boxes with confidence > 0.7 and then cropping the image to optimize resources. Then, the cropped

images will be passed through MediaPipe Pose to obtain the landmarks of the human pose by detecting 33 landmarks or any subset of them that consists of key points such as shoulders, hips, and knees by calling `mp_pose.Pose` with `min_detection_confidence` set to 0.5. The feature set considered is as follows: body aspect ratio (height/width > 2 , indicating the potential fall), height of the center of mass (normalized body height < 0.4), and joint angles (e.g., the angle formed by the hips and knees is less than 120°). Velocity is also measured as the function of the differences from frame to frame. This approach leverages the strengths of YOLO's detection capabilities with the precision offered by pose detection, hence minimizing the occurrence of false alarms through the consistency of the detection process with multiple frames while improving the interpretation of human pose dynamics.

4.9. Classification and Decision System

The multi-stage process begins with the use of the YOLO technique to identify the classes "person" and "fall" with their bounding boxes and confidence score. Once the confidence score is greater than 0.7, pose analysis is utilized to identify characteristics such as horizontal orientation (angle between the torso and vertical plane less than 30 degrees), low center of mass, and sudden changes in velocity. This process is then filtered through either rule-based logic or a lightweight classifier such as Random Forest to make the decision. Temporal smoothing is then realized by requiring that this process be confirmed over at least three consecutive frames. The output of this process will be the fall status, confidence level, and timestamp.

4.10. Evaluation Methodology

Accuracy is assessed via precision, recall, F1-score on test sets. Efficiency metrics include inference time (ms/frame), FPS, power, CPU temperature, and memory. Testing uses 15% held-out data and real simulations, comparing quantized vs. full-precision models against literature benchmarks.

5. Dataset

Three publicly available data sets are utilized for training and evaluation purposes. The data sets are KFall [8], CAUCAFall [9], and Fall Detection Dataset (FDD) [10]. The first data set, KFall [8], comprises 5,075 motion sequences performed by 32 subjects, including 2,729 activities of daily living (ADL) in 21 different categories and 2,346 falling events in 15 different categories. The second data set, CAUCAFall [9], comprises 10 subjects who performed five different types of falls and five different types of ADL in uncontrolled home environments with a focus on realistic occlusion and varying lighting conditions in the captured RGB video data. The FDD data set [10] comprises 191 videos captured with different camera angles, with a focus on indoor falls for comparative purposes.

6. Expected Outcomes

The following are the expected outcomes and evaluation criteria in this project:

Prototype Model Deployment: A fully integrated Raspberry Pi 5-based system for continuous elderly fall detection using overhead images, while at the same time sending out hazard detection signals to caregivers in critical areas such as stairs or living rooms.

High Reliable Detection: A customized, quantized neural network (adapted from MediaPipe Pose or YOLO-Pose) will achieve precision, recall, and F1-score of at least 95% on the detection of falls, near-falls, and immobility in elderly populations.

System Robustness Validation: The system will demonstrate consistency and good performance under challenging conditions such as different levels of lighting, partly obstructed areas, top-down camera angles, and changing movements such as climbing stairs, in which standard approaches fail to succeed.

Real-Time Processing Speed: The quantized model will allow for sustained performance of 15 to 20 FPS inference performance with Raspberry Pi 5 hardware and an approximation of 75% parameter reduction and low power consumption below an average of 5 watts.

7. Percentage of Image Processing & Computer Vision Involvement

The core of this mini project is strongly centered on computer vision and image processing techniques, which constitute approximately 85% of the total work. The primary intelligence and decision-making of this Elder Watch system rely entirely on visual analysis of video frames captured from the ceiling mounted camera. The key computer vision and image processing components include:

Real-time capture and preprocessing of video frames- This includes adjusting brightness and contrast, reducing noise, and correcting the camera perspective so that overhead views are properly aligned using OpenCV.

Human body keypoint detection- Extraction of 33 body landmarks in each frame using the pretrained MediaPipe Pose model. **Temporal movement and posture analysis-** Computation of keypoint displacement and velocity, torso inclination angle, vertical movement of head to feet, and relevant ratio to detect falls and unsafe conditions.

Deep learning model adaptation and optimization- Fine tuning of the pose model, followed by post-training quantization (INT8) using TensorFlow Lite to achieve real time performance on Raspberry Pi 5. **Fall decision logic-** Rule based or supervised classification of pose-motion sequences into categories such as normal activity, near fall, actual fall, or lying down too long. The remaining 15% covers supporting elements such as:

- Hardware integration (Raspberry Pi camera interface, systemd service setup),
- Alerting mechanism (SMS/email notification),
- Basic status display on the touchscreen.

All of these non-vision parts exist only to enable or report the results of the computer vision pipeline. Thus, they do not contribute to the actual hazard detection logic.

References

- [1] Q. Zou, Y. Wang, and Z. Chen, “Deep learning for computer vision-based activity recognition and fall detection of the elderly: A systematic review,” *Applied Intelligence*, 2024.
- [2] A. Giri, A. Hasib, and A. S. M. Akib, “Real-time human fall detection using yolov5 on raspberry pi: An edge ai solution for smart healthcare,” in *Lecture Notes in Networks and Systems*, vol. 1597, pp. 493–507, Springer, 2026.
- [3] E. Alam, A. Sufian, P. Dutta, and M. Leo, “Vision-based human fall detection systems using deep learning: A review,” *Computers in Biology and Medicine*, vol. 146, p. 105626, 2022.
- [4] A. Paramasivam and K. S. Shashikala, “Development of artificial intelligence edge computing based wearable device for fall detection and prevention of elderly people,” *Heliyon*, vol. 10, no. 8, p. e28688, 2024.
- [5] T. Riahi, M. A. H. Bappy, and M. M. Islam, “Elderfallguard: Real-time iot and computer vision-based fall detection system for elderly safety,” 2025.
- [6] J. Smith *et al.*, “Senior fall detection through vision and vibration techniques with raspberry pi,” *International Journal of High School Research (IJHSR)*, vol. 6, no. 2, pp. 113–120, 2024.
- [7] M. T. Alam, Y. Wang, and B. Yildirim, “Fast and high-precision human fall detection using improved yolov8 model,” *IEEE Access*, vol. 12, pp. 5271–5283, Sept. 2024.
- [8] KFall Dataset Team, “Kfall dataset.” <https://sites.google.com/view/kfalldataset>, 2022. Accessed: January 2026.
- [9] S. Giraldo-Forero *et al.*, “Caucafall: A multimodal dataset for fall detection in uncontrolled home environments,” *Data in Brief*, vol. 45, p. 108672, 2022.
- [10] Fall Detection Dataset Team, “Fall detection dataset (fdd).” <https://falldataset.com/>, various. Accessed: January 2026.